

technical software has been developed to cater to the needs of communities engaged in top-class research in science, engineering, and technology.

In this article, we will acquaint ourselves with one such category of software: interactive theorem provers.

Undoubtedly, all of us have had to prove some theorems in our lifetime, with the Pythagorean theorem being a prime example. But why do we need interactive theorem provers? Imagine that you are a researcher confident of proving an essential theorem. However, how can the research community fully trust your proof? There might be an unnoticed error in your reasoning. This is where peer review comes to our rescue. To gain credibility, you must submit your proof to a reputable journal, where experts in your specific area of research will evaluate your work and verify its accuracy and authenticity. However, human error can still creep in, and indeed, such issues have occurred in the research community on numerous occasions.

## The history of the four-colour theorem

Let us briefly go through the history of the famous four-colour theorem as an example of erroneous proofs being mistakenly accepted as correct. The four-colour theorem states that no more than four colours are required to colour the regions of any map so that no two adjacent regions sharing a boundary will have the same colour. Notice that if two or more regions meet only at a point, then they can share the same colour. Figure 1 shows an example of a map coloured in just four colours (*Courtesy: Wikipedia*). The observation was first made by Francis Guthrie in 1852. In 1879, a proof was proposed by Kempe. This proof was widely accepted as correct. However, in 1890, Heawood proved that Kempe's proof was incorrect. Notice that an incorrect proof was believed to be true for almost 11 years.

The work by Kempe was used by Heawood to prove a weaker theorem known as the five-colour theorem. After many false and incomplete proofs and almost a century later, the four-colour theorem was finally proved in 1976 by Appel and Haken. However, their proof was computer-based, and due to the immense size of this proof, its veracity was

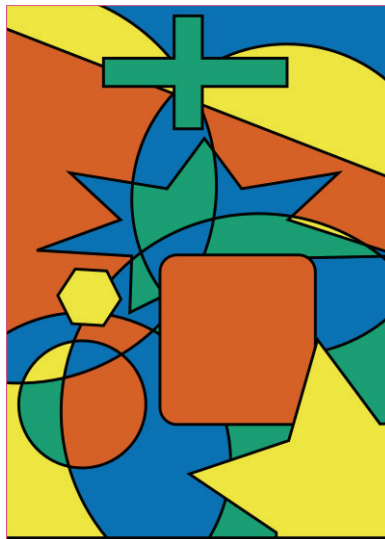


Figure 1: Example of a four-coloured map (*Courtesy: Wikipedia*)

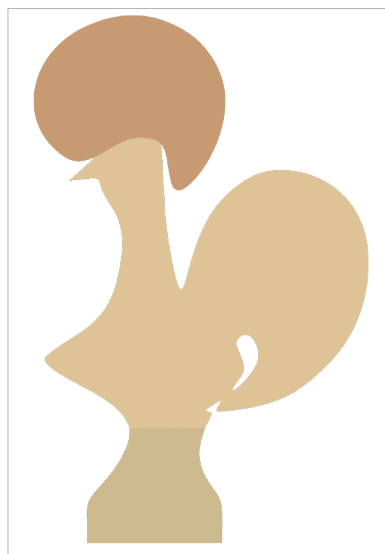


Figure 2: Logo of Coq

still doubted by many. A simpler proof using the same ideas (again relying on computers) was published in 1997 by Robertson, Sanders, Seymour, and Thomas. Finally, in 2005, the theorem was proved by Werner and Gonthier using Coq, thus ending any doubts about the proof of the theorem. Now, we can place our trust solely in the Coq kernel. So, in a nutshell, an interactive theorem prover can be a powerful tool in establishing the validity of complex theorems.

In this article, we will only introduce two interactive theorem provers: Coq and Lean. Although the article will not delve into the details of these two software, it will serve an essential purpose. The readers will become aware of the abundance of solutions offered by the open source software community, and they will also realise that solutions may exist even for the most difficult problems.

## Coq

Coq is free and open source software developed by INRIA, a French research institute focusing on computer science and applied mathematics. Coq operates under the LGPLv2.1 licence. It was first released in 1989, and the latest stable version is version 8.17.1. Figure 2 shows the logo of Coq. As mentioned earlier, a definitive proof of the four-colour theorem is one of the early success stories of Coq. The Feit–Thompson theorem is another important theorem proved using Coq. Coq was developed using Ocaml, which is a multi-paradigm programming language with emphasis on functional programming.

Now, let us see how to use Coq from our system. One good thing is that you don't need to install Coq on your system to work with it. You can run it in your browser using a web-based environment called jsCoq. This environment is available online at <https://coq.vercel.app>. Figure 3 shows the terminal of the jsCoq environment. For beginners, I suggest using this web-based environment.

However, Coq is also cross-platform software; it can be installed and used on Windows, macOS, and Linux. In Linux, Coq can be used with different interfaces like Vim, Emacs, etc. As an example, we will see how to install and use Coq in Ubuntu. I have used Proof General, an Emacs interface for Coq. Usually, Emacs is installed in Ubuntu by default;